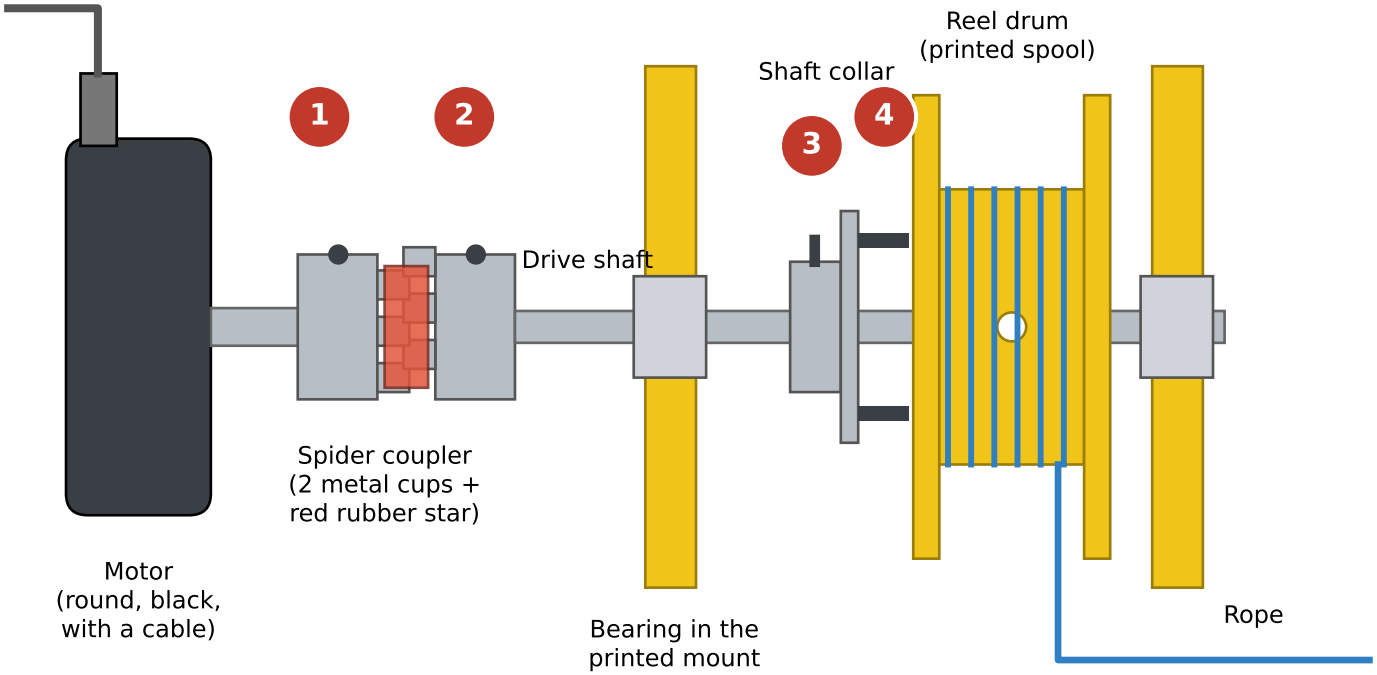
Motor unit: what turns the rope

Side view of one tower's motor unit (Stanford design; IBMS drums are printed from the same file). The motor turns the parts from left to right. Red numbers = the 4 places where it can slip.



Where it can slip

1	**Motor shaft → coupler**	the coupler's left cup turns on the motor shaft
2	**Coupler → drive shaft**	the coupler's right cup turns on the drive shaft
3	**Drive shaft → shaft collar**	the collar turns on the drive shaft (clamp screw loose)
4	**Shaft collar → reel drum**	the drum turns on the collar (screws loose or plastic torn)
–	**Rope → drum**	the rope slides on the drum (knot not held, too few turns)

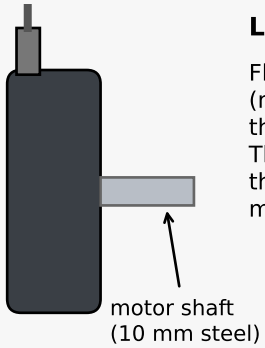
Your description (the drum stops when the rope gets tight, the motor speeds up) fits 3 or 4 best.

Bearings never hold the drum: they only let the shaft spin. Not a slip point.

SAFETY: disarm in the UI, then switch the tower's 48 V OFF before touching the motor unit.

The parts, one by one

Motor + motor shaft



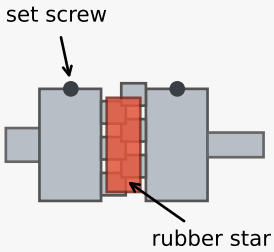
Looks like

Flat round motor (maxon EC 90 flat, 9 cm), thick cable on it. The short steel pin in the middle is the motor shaft.

Check

Nothing to tighten on the motor itself. It is where the marker line starts.

1–2 Spider coupler



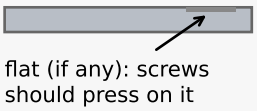
Looks like

Two metal cups facing each other, claws interlocking, a coloured rubber star between. Each cup has a small screw on its side (or a slot with a clamp screw).

Check

Joint 1 = left cup on the motor shaft.
Joint 2 = right cup on the drive shaft.
Tighten both screws (hex key).

Drive shaft



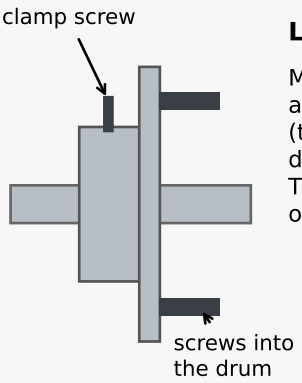
Looks like

Plain steel rod (about 9.5 mm thick) from the coupler, through the bearings in the printed mount, to the drum.

Check

If it has a flat side, the coupler and collar screws must press on the flat.

3–4 Flanged shaft collar



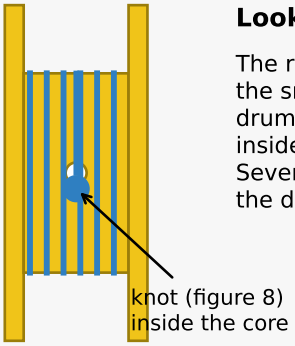
Looks like

Metal ring with a slit and one screw across it (the clamp), plus a flat disc with small holes. The disc is screwed onto the drum.

Check

Joint 3 = ring on the shaft:
tighten the clamp screw firmly.
Joint 4 = disc on the drum:
tighten the small screws
(torn plastic: see page 4).

Rope on the drum



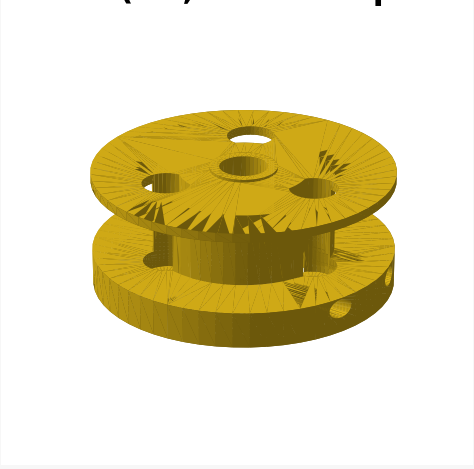
Looks like

The rope goes through the small hole in the drum core; a knot inside stops it. Several turns stay on the drum when tight.

Check

Knot holds? At least 3 turns left on the drum at full tension?

Reel drum (3D, from the print file)

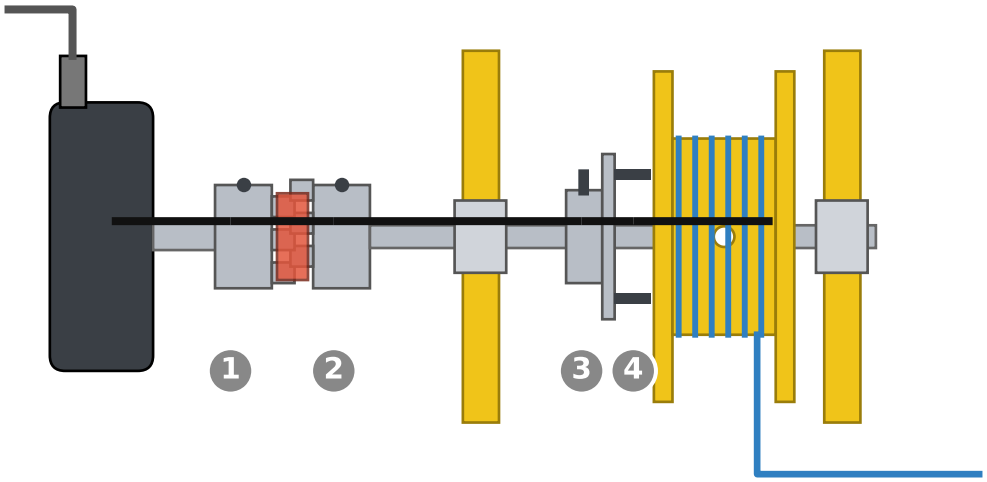


Printed spool: rope surface 38 mm, flanges 64 mm, tie hole 5.6 mm.

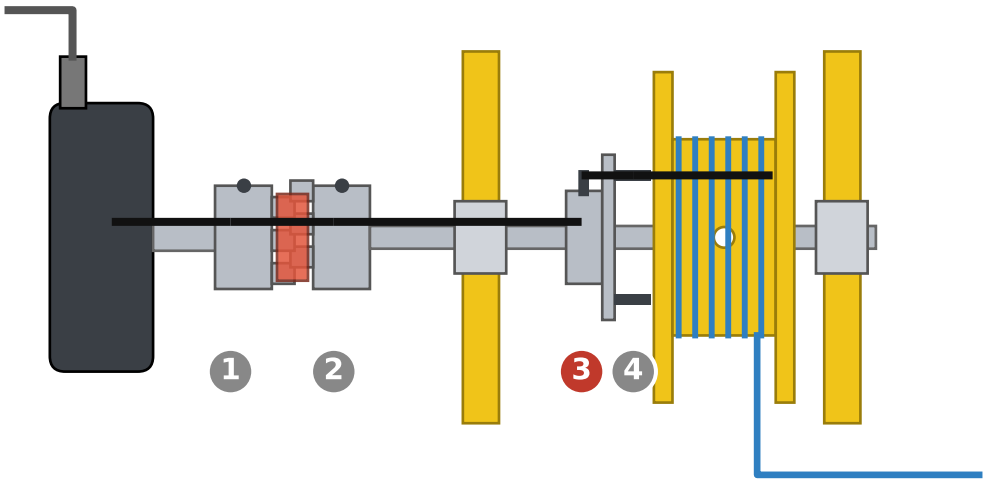
Find the slip: the marker-line test

1. 48 V OFF for this tower (disarm first).
2. Draw ONE straight line with a permanent marker along the motor shaft, both coupler cups, the drive shaft, the shaft collar and the drum (top picture).
3. Hold the drum firmly and turn the motor by hand, hard (or: hold the motor, turn the drum).
4. Look where the line no longer lines up: that joint slips (bottom picture: joint 3).

Before: one straight line



After: line broken at joint 3 (collar turned on the shaft)



Line breaks at …

1	**coupler cup on the motor shaft**	tighten the left cup's screw (on the flat if there is one)
2	**coupler cup on the drive shaft**	tighten the right cup's screw
3	**collar on the drive shaft**	tighten the collar's clamp screw firmly
4	**drum on the collar**	tighten the small screws; if they spin freely, the plastic is torn (page 4)

Line intact everywhere but the rope still slips?
Then the rope slides on the drum: check the knot and the turns.

If the drum's screws spin in the plastic

The print is torn where the collar screws go in. Options, easiest first:

- Longer screws all the way through the drum, with a nut and washer on the other side.
- Heat-set threaded inserts in the drum (soldering iron, a few minutes per insert).
- Reprint the drum: 100 % infill (or 6+ walls) around the centre, PETG or ABS, not PLA.

After any fix

- Redraw the marker line, power the tower, do one test pull (testing page, 30 N, 1 s).
- The line must still be straight afterwards. Then 50 N, then 100 N.

Record what you found

- Which tower, which joint, what you changed: tell Claude, it goes into docs/STATUS.md.
- A phone photo of the joint helps (docs/hardware-photos/).

Drawings follow Stanford's motor unit (docs/reference/stanford-cad/1) Motor Unit):
spider coupler 10 mm / 3/8 in, drive shaft in flanged bearings, flanged shaft collar on the drum.
If your tower looks different, send a photo and the guide gets updated.